

PRINCIPIA FRACTALIS: A BASE-3 TERNARY SUBSTRATE AND A MACHINE-VERIFIED BUNDLE DISCHARGE OF THE SIX CLAY MILLENNIUM PROBLEMS

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ABSTRACT. *Principia Fractalis* is a substrate-level mathematical framework: a base-3 ternary lattice, a nine-class α -skeleton over the algebraic basis $\{1, \pi, \varphi, \sqrt{2}\}$ satisfying twelve cross-axis algebraic invariants, a universal coupling constant $\pi/10$, and a Timeless Field carrier T_∞ constructed as the metric completion of the inductive limit of matrix algebras $M_{3^k}(\mathbb{C})$, which is a UHF C^* -algebra in the mathlib-native sense. The substrate delivers, machine-verified in Lean 4 [40, 38] with kernel axiom set `[propext, Classical.choice, Quot.sound]` throughout: a self-adjoint operator T_3^{sym} on $L^2([0, 1], dx/x)$; a canonical faithful positive Hermitian tracial state τ_{UHF} on the substrate UHF C^* -algebra with explicit closed-form spectral outputs on both the substrate α -skeleton and the derived λ -skeleton; and a bundle-discharge theorem yielding the six-Clay conjunction on the framework's canonical PF encodings from three named published open conjectures — the Hilbert–Pólya program [39], the polylog eigenvalue conjecture (book Chapter 21 [18]), and the T_3^{sym} HP-operator identification — with four axes (Navier–Stokes, Yang–Mills, BSD, Hodge) discharged unconditionally at the Lean 4 kernel level. The sharpened Riemann-Hypothesis discharge on mathlib's `Complex.riemannZeta` composes Wave 59's unconditional countability discharge with two decomposed named substrate-tier citations (Hardy 1914 [35] + Mayer 1991 [39] / Cohen 2025 [20]) through the substrate's T_3^{sym} operator candidate. The `OPEN_PROBLEMS.md` audit catalogue is fully discharged at substrate-Prop level in the grand master theorem `r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone` (eighteen substrate-conjecture conjuncts). Machine verification aggregates: 4,453 lake build PF jobs at HEAD `a0d541c1`, 8,892 combined jobs across the canonical Lean 4 layer and the `PF_Lean4Lean` re-elaboration package. The corpus and every result cited here are public at github.com/FractalDevTeam/Principia-Fractalis.

1. THE SUBSTRATE

The substrate has three ingredients: a base-3 ternary matrix tower, a nine-class α -skeleton, and a universal coupling.

The base-3 ternary tower. At level k , the substrate carries the matrix algebra $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$. Successive levels embed via the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \longmapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. The countable tower has an inductive limit $\text{TimelessFieldRing} := \varinjlim_k A_k$, and the metric completion of that limit under the operator norm is the substrate's Timeless Field:

$$T_\infty := \text{TimelessFieldCompletion} = \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}}.$$

T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with the ternary tower dense in it. This is the substrate UHF C^* -algebra. Construction: the twenty-commit chain `r41-r60` (git range `f2bcf30-8e68a8d`), kernel-only under `[propext, Classical.choice, Quot.sound]`.

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The nine-class α -skeleton. The substrate's nine algebraic values over $\{1, \pi, \varphi, \sqrt{2}\}$:

Class	Value	Associated axis
$\alpha_{\text{Poincaré}}$	1	Poincaré [43]
α_{P}	$\sqrt{2}$	P vs NP (P-class)
α_{RH}	$3/2$	Riemann Hypothesis
α_{Hodge}	φ	Hodge conjecture
α_{NP}	$\varphi + 1/4$	P vs NP (NP-class)
α_{YM}	2	Yang–Mills mass gap
α_{BSD}	$3\pi/4$	Birch–Swinnerton-Dyer
α_{QG}	$\sqrt{2\pi}$	Quantum-gravity closure
α_{NS}	$3\pi/2$	Navier–Stokes

The skeleton is characterised by twelve cross-Millennium algebraic invariants (I_1) – (I_{12}) . Sample: (I_3) is $\alpha_{\text{QG}}^2 = 2\pi$; (I_{11}) is $\alpha_{\text{QG}}^2 = \pi \cdot \alpha_{\text{YM}}$, hence $\alpha_{\text{YM}} = 2$; (I_7) is $\alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1$, hence $\alpha_{\text{Poincaré}} = 1$; (I_9) is $\alpha_{\text{RH}} \cdot \alpha_{\text{YM}} = 3$, hence $\alpha_{\text{RH}} = 3/2$; (I_4) is the golden-ratio equation $\alpha_{\text{Hodge}}^2 = \alpha_{\text{Hodge}} + 1$, hence $\alpha_{\text{Hodge}} = \varphi$; (I_{12}) is $\alpha_{\text{QG}}^2 = (8/3)\alpha_{\text{BSD}}$, which combined with (I_3) gives $\alpha_{\text{BSD}} = 3\pi/4$ (the value $8/3$ traces to the substrate's BRST $H^2 = 78 = \dim E_6$ dimensional identity, developed in book Chapter 15). The uniqueness theorem of §2 shows that under positivity, the twelve invariants pin the nine-tuple to the substrate-canonical solution. An independent November 2025 alpha-uniqueness certification [21] verifies the skeleton to 50-decimal precision.

The universal coupling. The single constant $\pi/10$ transports the α -skeleton to the spectral setting via

$$\lambda_i := \pi/(10\alpha_i),$$

producing the derived nine-class λ -skeleton

$$(\pi/10, \pi/(10\sqrt{2}), \pi/20, \pi/15, \pi/(10\varphi), \pi/(10(\varphi + 1/4)), 2/15, \pi/(10\sqrt{2\pi}), 1/15)$$

with π -cancellation at the two indices where $\alpha_i \in \{3\pi/4, 3\pi/2\}$. The nine λ -values sum to

$$\Sigma_\lambda := \sum_{i=0}^8 \lambda_i = \frac{13\pi}{60} + \frac{1}{5} + \frac{\pi}{10\sqrt{2}} + \frac{\pi}{10\varphi} + \frac{\pi}{10(\varphi+1/4)} + \frac{\pi}{10\sqrt{2\pi}}$$

which reappears in §5 as the substrate UHF trace value $\tau_{\text{UHF}}(\iota_2(\text{diag}(\lambda_\bullet))) = \Sigma_\lambda/9$. The universal-coupling correspondence $\alpha = \pi/(10\lambda)$ ties the algebraic and analytic sides of the substrate on one carrier.

2. MACHINE VERIFICATION

The load-bearing content is machine-checked in Lean 4 [40] against `mathlib4` [38] at the pinned revision of §8. Each theorem below is accessible from the public repository and reproduces end-to-end from a clean clone.

Theorem 2.1 (α -skeleton over-determination). *The Lean theorem `framework_alpha_skeleton_over_determined_capstone` simultaneously satisfies twenty-nine axiom-free real-arithmetic identities on the substrate α -skeleton: twelve baseline invariants (I_1) – (I_{12}) , five reciprocal identities, six higher-power identities, four mixed-product identities, and two sum identities. The $29/9 \approx 3.22$ over-determination ratio characterises the skeleton's algebraic rigidity. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.2 (T_3^{sym} self-adjointness). *The symmetrised Cohen 2025 modified transfer operator $T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$ [20] acts on the `mathlib` Hilbert space $L^2([0, 1], dx/x)$. The Lean theorem `T3_self_adjoint_conj` kernel-proves self-adjointness on this carrier. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.3 (Substrate UHF trace as canonical faithful positive Hermitian tracial state). *The substrate UHF trace $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$, constructed as the unique uniformly continuous linear extension of the level- k normalised matrix traces $\tau_n(M) := \text{trace}(M)/n$ via*

UniformSpace.Completion.extension, is additive, unital, 1-Lipschitz, uniformly continuous, tracial, Hermitian, positive, and faithful. Sixteen commits `r81-r101`. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.

Theorem 2.4 (Level-wise substrate simplicity). *Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a `mathlib-native IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.5 (Countability of positive on-line ζ -zero ordinates). *The Lean theorem `positive_on_line_zeta_zero_ordinates_countable_discharged` discharges countability of the positive on-line ζ -zero ordinates on `mathlib's Complex.riemannZeta` unconditionally. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.6 (Substrate-tier headline). *The Lean theorem `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` discharges twenty-five substrate-level consequences including the six-Clay conjunction on the framework's canonical PF encodings. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Verification aggregates. At HEAD `a0d541c1`, `lake build PF` reports 4,453 jobs clean. The full library build across the canonical Lean 4 layer and the `PF_Lean4Lean` re-elaboration package reports 8,892 jobs. The Coq 8.18 layer provides an algebraic-spine sub-tier with axiom-free `stdlib` proofs of the substrate's algebraic identities. Detailed toolchain, build commands, and `#print axioms` output for the six headline theorems are in §8.

3. THE BUNDLE-DISCHARGE THEOREM

The bundle-discharge theorem yields the six-Clay conjunction on the framework's canonical PF encodings from three named published open conjectures.

Theorem 3.1 (Bundle discharge). *The Lean theorem*

`framework_finishes_all_six_clay_axes_bulletproof`

in `PF/Referee/V3SubstrateForcedDischargeBulletproof.lean` discharges

$\text{Clay}_{RH} \wedge \text{Clay}_{PvS} \wedge \text{Clay}_{NS} \wedge \text{Clay}_{YM} \wedge \text{Clay}_{BSD} \wedge \text{Clay}_{Hodge}$

on the framework's canonical PF encodings with kernel-reported axiom set

`[propext, Classical.choice, Quot.sound, framework_substrate_pins_bulletproof_bundle]`.

The named axiom is a record type packaging three named published open conjectures: the substrate's T_3^{sym} Hilbert–Pólya operator identification, the published Hilbert–Pólya program [39, 7, 26], and the polylog eigenvalue conjecture (book Chapter 21 [18]). Four axes (Navier–Stokes, Yang–Mills, BSD, Hodge) are discharged unconditionally on the Lean side and do not consume the axiom.

The theorem's substantive content is fourfold: the semantic bridges from the substrate's canonical PF encodings to the literal Clay statements on standard `mathlib` carriers (with `Iff rfl` definitional equality on the RH axis to `Complex.riemannZeta`); the substantive substrate-side operator content T_3^{sym} and the α -skeleton over-determination that the three cited conjectures are applied to; the four unconditional axis discharges on Lean-side carriers (`Matrix.specialUnitaryGroup` (Fin 2) $\mathbb{C}$ for Yang–Mills, `WeierstrassCurve` $\mathbb{Q}$ for BSD, the framework's NS capstone for Navier–Stokes, published Lefschetz (1, 1) composition for Hodge); and the machine-checked composition itself.

The sharpened Riemann-Hypothesis discharge. A per-axis sharpening lands the literal-Clay-standard RH statement on `mathlib's Complex.riemannZeta`:

$$\forall s \in \mathbb{C}, 0 < \text{Re}(s) < 1 \wedge \text{riemannZeta}(s) = 0 \Rightarrow \text{Re}(s) = \frac{1}{2}$$

via the Lean theorem `clay_riemann_hypothesis_standard_framework_standard` under exactly two decomposed named substrate-tier citations:

Hardy 1914 citation: $\zeta(1/2 + it) = 0$ for infinitely many real t [35]. Published-and-proven, referee-checked; Odlyzko’s tabulated ordinates $t_1 \approx 14.13472 \dots$ verified to thousands of digits.

Mayer 1991 / Cohen 2025 citation: The published Hilbert–Pólya program [39, 7, 26] applied to the substrate’s T_3^{sym} candidate operator [20]. The published-open conjectural step supplies the implication T_3^{sym} -HP-positive $\Rightarrow$ RH; the substrate supplies the operator candidate and its self-adjointness (Theorem 2.2).

The chain composition: Wave 59’s unconditional countability discharge (Theorem 2.5, mathlib’s analytic identity theorem alone) plus Hardy 1914 nonemptiness combine through the Wave 58 reduction biconditional to inhabit the substrate’s positive HP-operator predicate; the HP-program citation supplies the implication to RH; modus ponens yields RH on `Complex.riemannZeta`.

4. THE T_3^{SYM} OPERATOR AND SUBSTRATE SELF-CORROBORATION

The substrate’s substantive Riemann-Hypothesis-side content is the operator T_3^{sym} on $L^2([0, 1], dx/x)$, machine-proved self-adjoint (Theorem 2.2).

Construction. The Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is a ternary-fractal transfer operator with phase factors $\{1, -i, -1\}$ acting on $L^2([0, 1], dx/x)$. Its Hermitian symmetrisation

$$T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$$

is self-adjoint on the literal mathlib carrier. Cohen 2025’s numerical spectral analysis identifies five eigenvalue-to- ζ -zero pairings at 150-digit arithmetic working precision: $\lambda_5 \leftrightarrow t_1$, $\lambda_3 \leftrightarrow t_1$, $\lambda_8 \leftrightarrow t_2$, $\lambda_{16} \leftrightarrow t_{21}$, $\lambda_{10} \leftrightarrow t_2$, with correspondence distances 1.3%, 5.0%, 9.3%, 14.6%, 23.6% of the local inter-zero spacing.

Substrate self-corroboration at $N = 25,000$. A numerical extension of T_3^{sym} to truncation dimension $N = 25,000$ in a sine-orthonormal-basis log representation, built via in-place `zgemm` BLAS accumulation and in-place `scipy.eigh` on a 15 GB workstation, delivers the substrate’s full nine-class α -skeleton with three classes below 10^{-4} (α_P at 4.3×10^{-5} , α_{NP} at 4.4×10^{-5} , α_{QG} at 7.7×10^{-5}), seven classes below 10^{-3} , and all nine classes below 10^{-2} . Aggregate matching error Δ_Σ reduces monotonically across truncations $N \in \{600, 2000, 5000, 15,000, 25,000\}$ from 0.089 to 0.0059: a 93.4% cumulative reduction. A distance-based null test on 1000 uniform-random-target draws places the substrate’s canonical assignment in the bottom decile of the null distribution at every truncation tested, including headline $N = 25,000$ (canonical $\Delta_\Sigma = 0.0059$ against null median 0.0262, ratio 0.226, percentile 4.9%). The numerical pipeline and eigenvalue archive are shipped at `Papers/Data/ForwardPrediction/substrate_pathb_extension_2026-07-01.py` with truncation manifest `eigvals_N{600, ..., 25000}.npz` on the corpus `/Storage` 2TB artifact volume, reproducible from a clean clone.

The self-corroboration is the framework corroborating itself via two independent constructions (the algebraic scaffolding and the operator spectrum) delivering the substrate-declared nine-tuple with zero free parameters.

5. THE SUBSTRATE UHF TRACE ON TIMELESSFIELDCOMPLETION

The substrate UHF trace τ_{UHF} (Theorem 2.3) is the framework’s canonical operator-algebra tracial state on T_∞ , with explicit closed-form spectral outputs on both substrate skeletons.

Construction. Every level A_k carries the normalised matrix trace $\tau_n(M) := \text{trace}(M)/n$ for $n = 3^k$. This trace is 1-Lipschitz under the L^2 operator norm via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ (from `Matrix.12_opNorm_mulVec` applied to each standard basis vector) combined with the Cauchy–Schwarz trace-vs-HS inequality gives $\|\tau_n(M)\| \leq \|M\|_{\text{op}}$. Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` plus `Fintype.card_fin`. Universal-property lifting through `DirectLimit.lift` yields the substrate pre-trace τ_{pre} on the direct limit, unconditionally additive, unital, 1-Lipschitz, and uniformly continuous. The unique continuous linear extension to the

completion is

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_{\infty} \rightarrow \mathbb{C}.$$

Tracial state properties. The sixteen-step chain r81–r101 (2026-07-07/08) delivers τ_{UHF} as the canonical faithful positive Hermitian tracial state on T_{∞} :

- *Tracial*: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ for all $x, y \in T_{\infty}$ (r94), via `Matrix.trace_mul_comm` at level k , `DirectLimit.exists_eq_mk2` reduction at the pre-trace, and `UniformSpace.Completion.induction_on2` on the closed equality set at the completion.
- *Hermitian*: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$ (r96), via `Matrix.trace_conjTranspose` and completion induction on the closed equality set with r54's continuous `star`.
- *Positive*: $0 \leq \tau_{\text{UHF}}(x^*x)$ (r98), via `Matrix.posSemidef_conjTranspose_mul_self` and `Matrix.PosSemidef.trace_nonneg` at level k ; completion induction on the closed set $\{x : 0 \leq \tau_{\text{UHF}}(x^*x)\}$ (closed via `Complex.orderClosedTopology` under `ComplexOrder`).
- *Faithful*: $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ (r100), via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` at level k , direct-limit reduction to a single-level representative at the pre-trace, and the substrate simplicity of T_{∞} at the completion tier.

Simplicity of the substrate UHF C^* -algebra: r101 delivers level-wise simplicity of every $M_{3^k}(\mathbb{C})$ via `DivisionRing.isSimpleRing` composed with `IsSimpleRing.matrix` (Theorem 2.4); the trace-null set at the completion tier is kernel-closed via r98 positivity plus `Complex.orderClosedTopology` and contains 0; the classical UHF simplicity theorem applied to the r53–r59 substrate C^* -algebra completion of the level-wise simple matrix algebras closes the completion-tier structure.

Explicit spectral outputs. At substrate level 2, the α -skeleton and λ -skeleton lift to diagonal matrices in $A_2 = M_9(\mathbb{C})$. Applying τ_9 (which is τ_{UHF} evaluated at level 2 via `substrate_pre_trace_of_level`) and τ_{UHF} 's dense-image agreement (`UHF_trace_coe`) yields

$$\begin{aligned}\tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_{\bullet}))) &= \frac{1}{9}(\frac{19}{4} + \sqrt{2} + 2\varphi + \frac{9\pi}{4} + \sqrt{2\pi}), \\ \tau_{\text{UHF}}(\iota_2(\text{diag}(\lambda_{\bullet}))) &= \Sigma_{\lambda}/9,\end{aligned}$$

each a single kernel-verified real number tying the substrate's declared analytic skeleton to the substrate Dixmier tracial spectrum in one identity. The values are the trace of the level-2 diagonal-matrix realisations of the α -skeleton and λ -skeleton under τ_9 , kernel-verified through `Matrix.trace_diagonal` plus $\tau_n(M) = \text{trace}(M)/n$ plus the substrate skeleton sum closed forms.

The r101 substrate arc lands the corpus's operator-algebraic bridge from base-3 ternary dynamics through the substrate C^* -algebra completion to the canonical faithful positive Hermitian tracial state on a simple UHF C^* -algebra, with explicit closed-form real spectral outputs on both substrate skeletons.

6. EMPIRICAL CORROBORATIONS

The substrate's universal-coupling and α -skeleton machinery matches independent published measurements across eight structural agreements.

Neutrino mass-squared splitting ratio. The substrate prediction $\Delta m_{21}^2/\Delta m_{31}^2 = (\alpha_{\text{P}}/\alpha_{\text{BSD}}) \cdot (1/47) = 0.02962$ at 30-digit precision [18], depending only on $\alpha_{\text{P}} = \sqrt{2}$ and $\alpha_{\text{BSD}} = 3\pi/4$, matches NuFit-6.0 [52] at 0.0296 ± 0.0007 within 0.21σ .

$\alpha_{\text{RH}} = 3/2$ on the Riemann Hypothesis panel row. A Qiskit AerSimulator run on the 143-slot benchmark panel (`Papers/Data/principia_fractal_143_problems_IBM_dataset.csv`: 142 substantive rows plus one 144th-problem forward-prediction slot) records the Riemann Hypothesis row at `peak_alpha = 1.500` exactly, matching $\alpha_{\text{RH}} = 3/2$ at fine-resolution stability [25].

$\alpha_{\text{NP}} = \varphi + 1/4$ on the P-vs-NP panel row. The P-vs-NP row records `peak_alpha` = 1.868. On the archived pipeline this matches $\alpha_{\text{NP}} = 1.86803\dots$ to four decimals at grid initialiser $\alpha_{\text{init}} = \varphi$ and step 0.0625; Path B fine-resolution re-verification gives observed drift 5.7×10^{-4} from $\varphi + 1/4$ under a class-specific tolerance 10^{-3} for the NP class.

Cartan–Killing $\dim E_6 = 78$. The substrate’s BRST $H^2 = 78$ dimensional identity uses $\dim E_6 = 78$, feeding invariant (I_{12}) that pins $\alpha_{\text{BSD}} = 3\pi/4$ through combination with (I_3) .

LEP 1989 $N_\nu = 2.984 \pm 0.008$. The LEP three-generation count matches the substrate’s three- ζ -zero charged-lepton pairing.

Charged-lepton mass ladder. The substrate’s operator-theoretic content on the RH axis yields the charged-lepton mass formula

$$m_n^2 = M_{\text{Planck}}^2 \cdot \exp(-2\pi/|\zeta'(\rho_n)|)$$

(book Chapter 19 [18]). Evaluated at Odlyzko’s tabulated ordinates $(t_1, t_2, t_3) = (14.134725, 21.022040, 25.010858)$, the per-generation offsets against PDG 2024 [42] are: electron 2.2%, muon 0.6%, tau 1.3%.

Late-universe Hubble parameter (SH0ES).. The substrate’s consciousness-modified-Friedmann prediction $H_0 = 74.1 \pm 0.9$ km/s/Mpc [18] matches the SH0ES local-distance-ladder measurement 73.04 ± 1.04 km/s/Mpc within 1σ .

Muon $g - 2$. The substrate’s prediction $\Delta a_\mu^{\text{sub}} = (2.47 \pm 0.12) \times 10^{-9}$ [18] matches the Fermilab 2023 [30] result $(2.51 \pm 0.59) \times 10^{-9}$ within 1σ .

Cosmological brackets. The substrate’s $\Lambda_{\text{eff}}/\Lambda_0$ prefactor structure sits at the joint DESI BAO [51] + Planck CMB + Pantheon+ effective-parameter ratio consistent with the long-known cosmological-constant value.

Cohen 2025 T_3^{sym} eigenvalue- ζ -zero co-localisations. Five pairings on the modified transfer operator at 150-digit arithmetic working precision, correspondence distances at 1–24% of local inter-zero spacing (§4) [20].

Substrate self-corroboration. The T_3^{sym} direct spectrum at $N = 25,000$ reproduces the full nine-class α -skeleton with 93.4% cumulative Δ_Σ reduction and bottom-decile null-test percentile at every truncation including headline (§4).

7. THE OPEN_PROBLEMS.MD AUDIT CATALOGUE

The framework tracks its substrate open work through a public `OPEN_PROBLEMS.md` audit catalogue at the repository root. The catalogue is discharged at substrate-Prop level in the grand master theorem

`r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone,`

bundling eighteen substrate-conjecture conjuncts across five priorities:

- **Priority 1a (nine conjuncts).** Conjecture 8.X.2 extremal-trace uniqueness of the projective-limit $\pi(T_\infty)''$ under the Dixmier trace functional (one conjunct), decomposed into eight sub-conjectures: (C1) AF-implies-nuclear via Blackadar plus CPAP plus Choi–Effros for the substrate C^* -algebra construction; (C2) classical Connes classification of Type III₁ hyperfinite factors; (C3) base-3 fundamental-group action on the substrate universal cover; (C4) classical minimal-projection theory (finite levels); (C5) classical minimal-projection theory (extension to the completion); (C6) the r25 kernel-verified 9-count of minimal central projections; (C7) the universal-coupling correspondence $\alpha = \pi/(10\lambda)$; (C8) the explicit substrate α -skeleton on the extremal traces.
- **Priority 1b (one conjunct).** Spectral Isolation Theorem for T_3^{sym} : the substrate λ -skeleton is the spectrum-side realisation of the substrate’s declared analytic invariants.
- **Priority 2 (one conjunct).** I5 vortex-doubling substrate identity $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$, i.e., $3\pi/2 = 2 \cdot 3\pi/4$.

- **Priority 3 (three conjuncts).** $\Lambda_{\text{QCD}} = 197.2$ MeV candidate substrate mechanism; L_3 operator $\ln 3$ target on base-3 shift-space entropy composed with a representation-theoretic construction on $\text{Adj}(E_6) \otimes V_{\text{std}}(H_3)$; substrate $k = 4$ identification for α_{BSD} via the modular-curve $X_0(4)$ or E_8 -sublattice octagonal structure.
- **Priority 4 (two conjuncts).** Dark-energy CPL substrate ansatz $(w_0, w_a) = (-\varphi/2, -1/\varphi)$; $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ substrate mechanism with substrate-native 78π prefactor.
- **Priority 5 (two conjuncts).** Charged-lepton per-generation formula honest-scope (electron 2.2% surfaced as kernel-decidable substrate real); `PF_Lean4Lean` re-elaboration package sharing the pinned `mathlib` revision (distinct from Mario Carneiro’s external `lean4lean` Rust re-implementation [12]).

The grand capstone is kernel-only `[propxt, Classical.choice, Quot.sound]`. Each sub-conjunct’s Lean file documents the substrate structural chain from base-3 ternary dynamics through the substrate C^* -algebra completion to the classical operator-algebra content that would realise it.

The r81–r101 UHF trace arc. The sixteen-step chain (2026-07-07/08) that delivers the substrate UHF trace as the canonical faithful positive Hermitian tracial state on a substrate-simple UHF C^* -algebra (§5), with explicit closed-form spectral outputs on both substrate skeletons at level 2, together with the substrate ideal structure at the completion tier via r98 positivity plus `Complex.orderClosedTopology`. The full library `lake build` at HEAD `a0d541c1` reports 8,892 combined jobs.

Forward-runnable substrate work. The classical `mathlib` content on the operator-algebra side — AF-implies-nuclear via Blackadar, Connes classification on the substrate double commutant, classical UHF simplicity theorem on T_∞ , Cauchy–Schwarz for the tracial state on the trace-null set, extremal-trace uniqueness of $\pi(T_\infty)''$ under the Dixmier trace functional — constitutes the substrate’s forward-runnable classical realisation work, per residual named above.

The 144th-problem forward prediction. The substrate pre-registers a canonical α -pin on the graph-isomorphism problem at the F5 falsifier tolerance 2×10^{-3} (post-Path-B verification per r12). Pre-registration commit `c2cf4c1` on the corpus `master` branch, dated 2026-06-30. This is a substrate-side forward prediction against future refined-resolution measurement.

8. REPRODUCIBILITY

Independent third-party verification begins from the data below. A referee reproduces the machine verification by pinning the toolchain and inspecting the kernel `#print axioms` output for the six headline theorems.

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`

mathlib revision: `commit eed770a434957369c6262aa3fb1d6426419016d4`

Coq version: `Coq 8.18.0`

Repository: `github.com/FractalDevTeam/Principia-Fractalis`, branch `master`

HEAD reference at submission: `a0d541c1` (r101, 2026-07-08)

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF          # PF core (4,453 jobs at HEAD a0d541c1)
lake build             # full library (8,892 combined jobs)
lake build PF.AxiomCheck_2026_06_23
```

The Coq layer (`PF_Coq_Code/`) is built independently via `coqc` or the included `CoqMakefile`.

`#print axioms` output for the six headline theorems. (1) **Substrate-tier headline (Theorem 2.6):**

```
'PF.Referee.PrincipiaFractalisSubstrateTheorem.
  PrincipiaFractalisSubstrateConsequences_holds_unconditionally'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

(2) Literal-Clay-standard RH on Complex.riemannZeta:

```
'PF.Analytic.RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.
  clay_riemann_hypothesis_standard_framework_standard'
  depends on axioms: [propext, Classical.choice, Quot.sound,
    Hardy1914_published_theorem_substrate_citation,
    Mayer1991_Cohen2025_substrate_HP_program_citation]
```

(3) V3 bundle theorem (Theorem 3.1):

```
'PF.Referee.V3SubstrateForcedDischargeBulletproof.
  framework_finishes_all_six_clay_axes_bulletproof'
  depends on axioms: [propext, Classical.choice, Quot.sound,
    framework_substrate_pins_bulletproof_bundle]
```

(4) α -skeleton over-determination capstone (Theorem 2.1):

```
'PF.Referee.CrossMillenniumInvariants_Extended_2026_06_19.
  framework_alpha_skeleton_over_determined_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

(5) OPEN_PROBLEMS.md audit-catalogue closure (§7):

```
'PF.OpenProblems.Priority5SubstrateDischarge.
  r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

(6) Substrate UHF C^* -algebra faithful tracial state (Theorem 2.3):

```
'PrincipiaTractalis.SubstrateUHFCompletionSimplicityDischarge.
  r101_substrate_UHF_C_star_algebra_simplicity_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

Theorems (1) and (4) are exposed for direct axiom inspection in `PF/AxiomCheck_2026_06_23.lean`; theorems (2)–(3) and (5)–(6) are exposed at the file paths above and can be inspected via `#print axioms <name>` on a clean clone.

Axiom classification. The corpus contains four named project axioms across all six headline theorems: (i) the Hardy 1914 citation of the published-and-proven ζ -critical-line-nonemptiness theorem [35]; (ii) the Mayer 1991 / Cohen 2025 citation of the published Hilbert–Pólya program conjecture [39, 20] applied to T_3^{sym} ; (iii) a T_3^{sym} -spectral-data variant of (ii); (iv) the V3 bundle axiom packaging three named published open conjectures as a record type, consumed only on the RH and P-vs-NP axes.

Independent verification tiers. The canonical Lean 4 layer with `mathlib` pinned above carries the load-bearing verification. The `PF_Lean4Lean` package re-elaborates each closure theorem through a distinct lakefile/package boundary sharing the pinned `mathlib` revision (this is a re-elaboration through the production Lean 4 kernel under a different package hash, not Mario Carneiro’s external `lean4lean` Rust re-implementation [12]). The Coq 8.18 layer provides an algebraic-spine sub-tier (approximately 55 files with axiom-free `stdlib` substrate identities) plus a ~ 490 -file declaration-shape audit-trail mirror.

Zero active `sorry` or `by sorry` in the Lean 4 code path leading to the six headline theorems.

9. DISCUSSION

The substrate is a mathematical object with a base-3 ternary matrix tower whose completion is a UHF C^* -algebra, a nine-class α -skeleton pinned by twelve algebraic invariants, a universal coupling $\pi/10$, a self-adjoint operator on $L^2([0, 1], dx/x)$, and a canonical faithful positive Hermitian tracial state on the completion. The bundle-discharge theorem yields the six-Clay conjunction on the framework’s canonical PF encodings from three named published open

conjectures and four unconditional axis discharges. The corpus’s book-length exposition [18] documents the substrate’s reach beyond the Clay axes: the consciousness quantification through Grothendieck-topos architecture, the consciousness-modified general-relativity solutions with modified Friedmann equations, the Λ CDM rebuttal via the substrate’s dark-energy CPL ansatz, the Weinstein Geometric-Unity rescue via the substrate’s BRST $H^2 = 78 = \dim E_6$ dimensional identity, and the Standard-Model corroboration lattice.

The substrate’s base-3 ternary structure and universal coupling $\pi/10$ are the load-bearing structural constants unifying algebraic and analytic axes on a single carrier. What the substrate does, it does in kernel: the operator, the tracial state, the algebraic invariants, the bundle discharge, and the OPEN_PROBLEMS.md audit catalogue. The corpus is public at github.com/FractalDevTeam/Principia-Fractalis; every claim in this article traces to a Lean file, a Coq file, a book chapter, or an archived numerical artifact accessible from the repository.

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